

Electric Potential

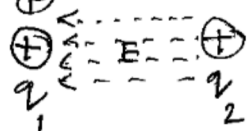
Chapter 24, R.H.W. - 10th Edition

Electric Potential

We identify basic force electric force we discussed before.
 → Our goal is to determine the force is conservative —
 " Whether a potential energy can be associated with it.

→ The motivation for associating a potential energy with a force is that we can then apply the principle of the conservation of mechanical energy to closed systems involving the force.

NOTE: Electric force is conservative and thus has an associated electric potential energy.

 → particle 1 is located at point P in the electric field of particle 2.

To account for the force between q_1 and q_2 we define an electric field $\vec{E}$.

particle or charge q_1 feel Electric field $\vec{E}$ setup by charge q_2 at point P. (pre-existing electric field due to q_2)

If we release (q_1) charge particle in pre-setup electric field E due to charge particle q_2 , (q_1) charge particle start to move and thus has kinetic energy.

→ Energy can not appear due to charge movement or charge movement need energy. Where does it come from?

We need to define a new concept electric potential V (a scalar quantity) associated with electric potential energy U . → Chapter 6

① Define electric potential ② How to calculate V , in different charge orientation, ③ Discuss how electric potential V is related to electric potential energy U .

Electric potential and Electric potential Energy.

(2)

First define Electric potential look back $\rightarrow$

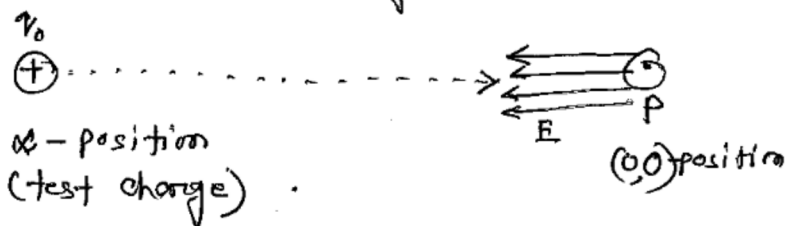
How ~~de~~ we define gravitational potential energy U of an object. first, we assign a position where we keep $U=0$. Second, we calculate work ' W ' the gravitational force does — if the object is moved up or down from that level. We then defined the potential energy as being $\boxed{U = -W}$ potential energy.

Let's follow the same procedure for Electric potential.

1 $\odot$ We need a reference point (configuration) where $U=0$.

2 $\odot$ The reasonable configuration is 'a test charge' far away (infinitely away) from the source of Electric field), at point 'p'

3 $\odot$ Next we bring this test charge in from infinity to the point p



so $W_{\infty \rightarrow 0}$ is the work done by the electric force. The work and thus the potential energy can be positive or negative depending on the sign of the charge at point p.

so, Electric potential becomes

$$\boxed{V = \frac{-W_{\infty \rightarrow 0}}{q_0} = \frac{U}{q_0}}$$

Electric potential is the amount of electric potential energy per unit charge, when a positive test charge is brought in from infinity. — ①

Units: Electric potential — represented with a special unit the volt (V). 1 volt = 1 joule per coulomb.

Electric field unit in terms of volt (V)

$$1 \text{ N/C} = \left(1 \frac{\text{N}}{\text{C}}\right) \left(\frac{1 \text{ V}}{1 \text{ J/C}}\right) \left(\frac{1 \text{ J}}{1 \text{ N.m}}\right) = 1 \frac{\text{V}}{\text{m}}$$

↳ definition of volt

Motion through an Electric Field

If we move from an initial point i to a second point f in the electric field of a charged object, the electric potential changed by $\Delta V = V_f - V_i$ — (I)

If we move a particle with charge q from i to f , then — potential energy of the system changed by

$$\Delta U = q \Delta V = q(V_f - V_i) \quad \text{--- (II)}$$

The change can be positive or negative, depending on the sign of q and ΔV . Zero, if there is no change in potential from i to f (same potential).

Because, the electric force is conservative, the change in potential energy ΔU between i and f is the same for all paths between these points.

Work by the field

$$W = -\Delta U \quad (\text{work, conservative force})$$

$$= -q(V_f - V_i)$$

Work attributed to force, W is the work done on the particle done by electric field. Work can be positive, negative, zero

① Conservation of Energy.

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If a charged particle moves through an electric field with no force acting on it, other than the electric force due to the field, then the mechanical energy is conserved.

Assume, we assign the electric potential energy to a particle. Conservation of Mechanical energy of the particle that moves from point i to point f as

$$U_i + K_i = U_f + K_f$$

$$\Delta K = -\Delta U$$

$$\boxed{\Delta K = -q \Delta V = -q(V_f - V_i)}$$

Work by an Applied force

If some force in addition to the electric force acts on the particle, we say that the additional force is an applied force or external force. — Such an applied force can do work on the particle, but the force may not be conservative and thus, in general — we cannot associate a potential energy with it.

(initial energy) + (work by applied force) = (final energy)

$$U_i + K_i + W_{app} = U_f + K_f$$

$$\Delta K = -\Delta U + W_{app}$$

$$\Delta K = -q \Delta V + W_{app}$$

$$W_{app} = q \Delta V \quad (\text{for } K_i = K_f)$$

— (1)

Sample problem 24.01 Work and potential energy in an electric field.1

Electrons are continually being knocked out of air molecules in the atmosphere by cosmic-ray particles coming in from space.

— Once electron released experiences an electric force $\vec{F}$ due to the electric field $\vec{E}$ that is produced in the atmosphere by charged particles already on earth.

— Electric field magnitude near earth surface

$E = 150 \text{ N/C}$ and is directed downward.

What is the change ΔU in the electric potential energy of a released electron when the electric force causes it to move vertically upward through a distance $d = 520 \text{ m}$. Through what potential change does the electron move?

$$\textcircled{1} \quad W = -\Delta U$$

$$W = \vec{F} \cdot \vec{d}$$

$$F = qE$$

$$W = q\vec{E} \cdot \vec{d}$$

$$= qEd \cos \theta$$

$$= qEd \cos 180 = -1.2 \times 10^{-14} \text{ J}$$

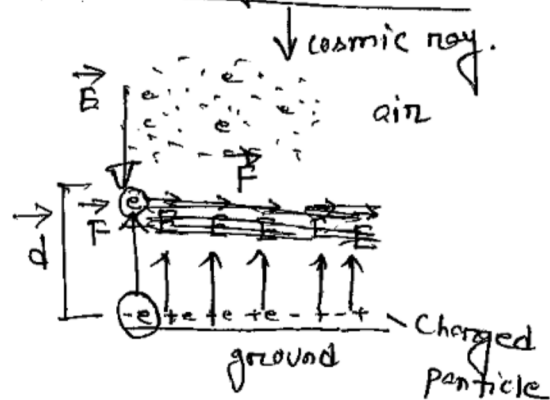
$$\Delta U = -1.2 \times 10^{-14} \text{ J}$$

During 520 m ascent, the electric potential energy of the electron decreased by $1.2 \times 10^{-14} \text{ J}$

$$\text{Electric potential: } \Delta V = \frac{\Delta U}{-q} = \frac{-1.2 \times 10^{-14} \text{ J}}{-1.6 \times 10^{-19} \text{ C}} = 4.5 \times 10^4 \text{ V}$$

$$= 45 \text{ kV}$$

This tells electric force does work to move the electron to a higher potential.



Cosmic ray, push the electron from air to ground.

Equipotential Surface and the Electric field.

⑤

Equipotential Surfaces

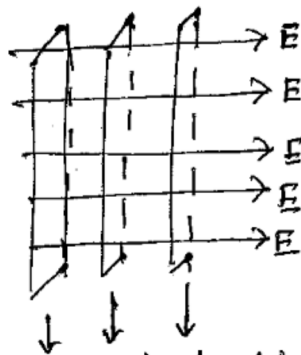
Adjacent points that have the same electric potential form an equipotential surface,

which can be — imaginary, real, or physical surface.

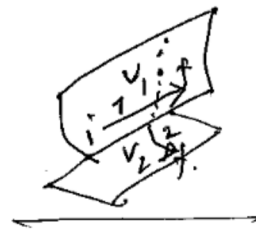
① No, net work W is done on a charged particle by an electric field, when the particle moves between two points i and f on the same equipotential surface.

W must be zero when $V_f = V_i$

② It means work is path independent on a given equipotential surface regardless of whether that path lies entirely on that surface.



equipotential surface.



Which path
work done is
zero.

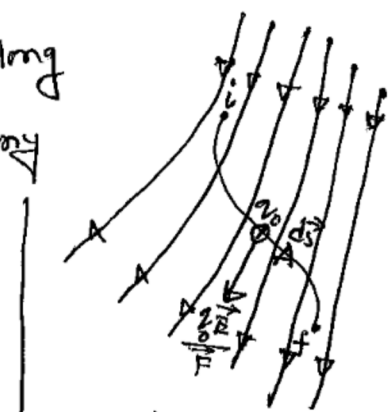
Calculating the Potential from the field

A positive test charge q , moves along the path from i to point f . At any

$$dW = \vec{F} \cdot d\vec{s}$$

$$dW = q_0 \vec{E} \cdot d\vec{s}$$

$$W = q_0 \int \vec{E} \cdot d\vec{s}$$



$\rightarrow q_0 \vec{E}$ electric force

~~Point~~ $V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$ $[W = -q(V_f - V_i)]$ act on the test charge and moves through a differential displacement.

Thus, the potential difference $V_f - V_i$ between any two points i and f in an electric field is equal to the negative of the line integral.

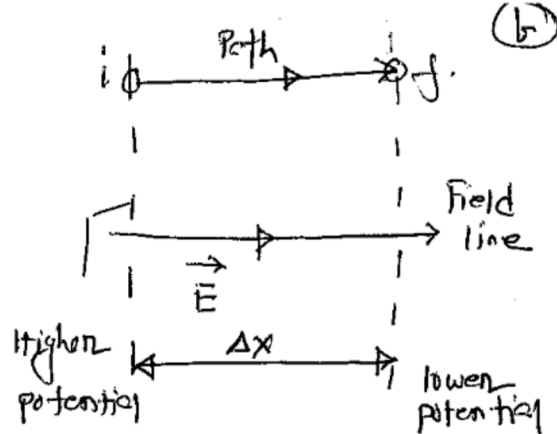
(*) As electric force is conservative, all paths yield the same results.

If we set $V_i = 0$, means we start from a zero potential and reach the potential with $V_f = V$, we can calculate potential at ^{that} point $V = - \int_i^f \vec{E} \cdot d\vec{s}$ where is this zero potential level.

We consider it at infinity $\rightarrow \infty$

Uniform field

We start a point i on an equipotential line with potential V_i and move to point f on an equipotential line with a lower potential V_f .



The separation between the two equipotential line is Δx . The angle between $\vec{E}$ and $d\vec{s}$ is zero.

$$\vec{E} \cdot d\vec{s} = E ds \cos 0 = E ds \quad [E \text{ is constant for a uniform field}]$$

$$V_f - V_i = -E \int_i^f ds \quad \text{--- (1)}$$

The integral is simply an instruction for us to add all the displacement elements ds from i to f .

$$\Delta V = -E \Delta x \quad (\text{uniform field})$$

This is the change in voltage ΔV between two equipotential lines in a uniform field, of magnitude E , separated by distance Δx .

Sample Problem 24.02: Finding the potential change from the electric field.

Potential due to a charged particle

$$\vec{E} \cdot d\vec{s} = E \cos \theta ds$$

① The electric field $\vec{E}$ is directed radially outward from the fixed particle. through P to infinity

$$\vec{E} \cdot d\vec{s} = E \cos \theta ds$$

$$V_f - V_i = - \int_R^\infty E dr$$

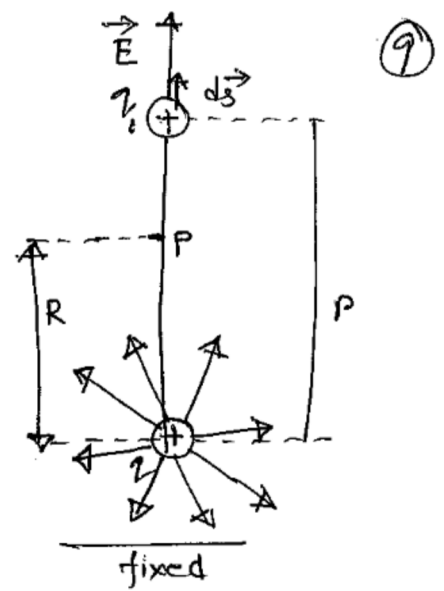
$$| V_f = 0 \text{ at } \infty \text{ and } V_i = V(R) |$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$0 - V = - \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_R^\infty$$

$$V = - \frac{1}{4\pi\epsilon_0} \frac{q}{R} \Rightarrow V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad [\text{switching } R \text{ to } r]$$



⊗ A positively charged particle produces a positive electric potential. A negatively charged particle produces a negative electric potential.

Potential Due to a Group of Charged Particles

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i} \quad (n \text{ charged particles})$$

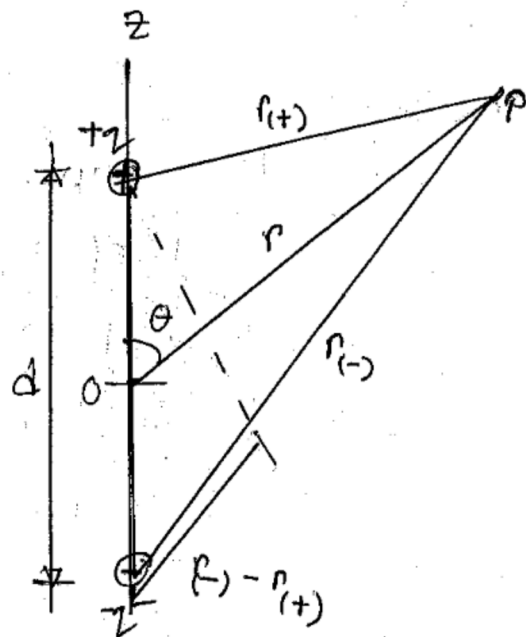
Algebraic sum.

Electric field $\rightarrow$ vector sum.

Potential Due to an Electric Dipole

At P, positively charged particle set up potential $V_{(+)}$ and the negatively charged particle sets up potential $V_{(-)}$. Then the net potential at P is given by

$$\begin{aligned}
 V &= \sum_{i=1}^2 V_i = V_{(+)} + V_{(-)} \\
 &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}} \right) \\
 &= \frac{q}{4\pi\epsilon_0} \frac{r_{(-)} - r_{(+)}}{r_{(-)} r_{(+)}} \quad \text{--- (1)}
 \end{aligned}$$



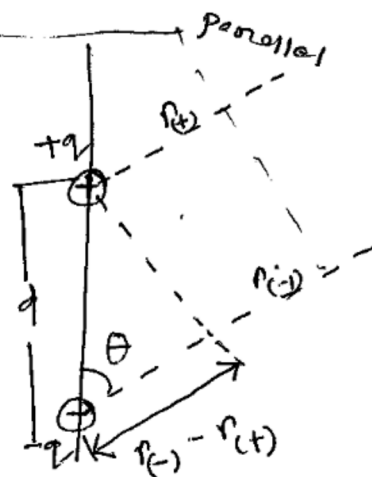
Naturally occurring dipoles. — Such as those possessed by many molecules — are quite small; So we are usually interested only in points that are relatively far from the dipole, such that $r \gg d$, where d is the distance between the charges and r is the distance from the dipole's midpoint to P.

$$r_{(-)} - r_{(+)} \approx d \cos \theta \quad \text{and} \quad r_{(-)} r_{(+)} \approx r^2$$

$$V = \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad \left[p = qd \right]$$

dipole moment



Thus, θ is measured from the direction of $\vec{p}$. We use this vector to report the orientation of an electric dipole.

Potential Due to a Continuous Charge Distribution

* When a charge distribution 'q' is continuous we can't

use $\rightarrow$ $\textcircled{*} V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$ (n charged particles)

we go for

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} \text{ (positive or negative } dq\text{)}$$

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \text{ --- } \textcircled{1}$$

Integral must be taken over the entire charge distribution. $\textcircled{*}$ Electric potential is scalar so no need to worried about any vector component.

Examine A line of charge and A disk charge to compute V, Section (24.5)

Calculating the Field from the Potential

* Suppose a positive test charge q_0 moves through a displacement $d\vec{s}$ from one equipotential surface to the adjacent surface. The work the electric field does on the test charge during the move is $\boxed{-q_0 dV}$.

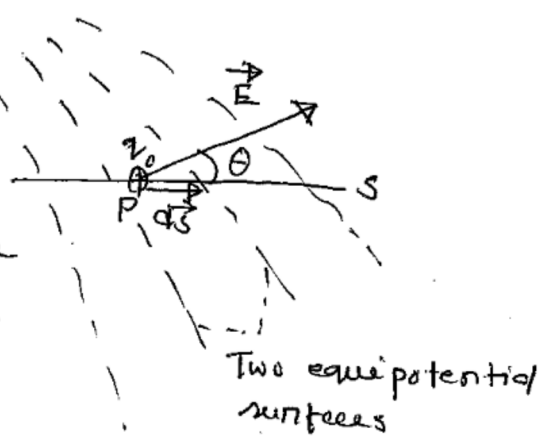
work done by the electric field may written as $\rightarrow (q_0 \vec{E}) \cdot d\vec{s}$ or $\boxed{q_0 E (\cos\theta) ds}$

$$-q_0 dV = q_0 E (\cos\theta) ds$$

$$E \cos\theta = - \frac{dV}{ds}$$

$$E_s = - \frac{\partial V}{\partial s} \quad \left[E_s \text{ is the component of } E \text{ along } \vec{s} \right]$$

$$E_x = - \frac{\partial V}{\partial x}, E_y = - \frac{\partial V}{\partial y}, E_z = - \frac{\partial V}{\partial z} \text{ --- } \textcircled{1}$$

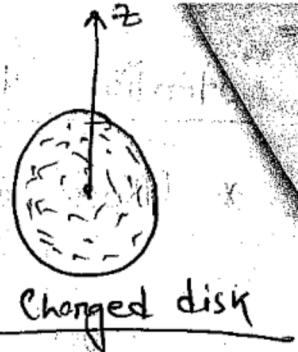


$\textcircled{*}$ For uniform electric field

$$\boxed{E = - \frac{\Delta V}{\Delta s}}$$

* Finding the field from the potential

The electric potential at any point on the central axis of a uniformly charged disk is given by,



$$V = \frac{\sigma}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$$

With this equation, derive an expression for the electric field at any point on the axis of the disk?

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\sigma}{2\epsilon_0} \frac{d}{dz} (\sqrt{z^2 + R^2} - z)$$

$$= \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

This is the same expression that is in Module 27-5.

Home work: verify it!

Calculating the field from the potential

Suppose a positive test charge q moves through a displacement $d\vec{s}$ from one equipotential surface to the adjacent surface. The work the electric field does on the test charge during the move is $-q dV$.

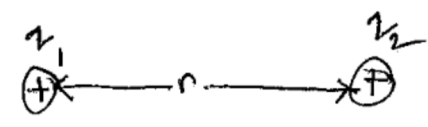
Work done by the electric field is $W = \int \vec{F} \cdot d\vec{s} = q \int \vec{E} \cdot d\vec{s}$

Since $W = -q \Delta V$, we have $\int \vec{E} \cdot d\vec{s} = -\Delta V$

For uniform electric field $\vec{E} = E \hat{s}$, $\int \vec{E} \cdot d\vec{s} = E \Delta s = -\Delta V$

So $E = -\frac{\Delta V}{\Delta s}$

Electric Potential Energy of a System of Charged Particles

System of two charged particles $\rightarrow$ 

1. Particle 2 fixed in place and particle 1 infinitely far away, with an initial potential energy U_i for the two-particle system.

2. Bring particle 1 to its final position, and then the system's potential energy is U_f . Our work changes the system's potential energy by $\Delta U = U_f - U_i$

$\rightarrow$ We can relate ΔU to the change in potential through which we move particle 1;

$$U_f - U_i = q_1 (V_f - V_i) \quad \text{--- (1)} \quad U_i = 0, \text{ as } r \rightarrow \infty$$

$V = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r}$, this tells us that when particle 1

is initially at distance $r = \infty$, the potential at its location is $V_i = 0$. When we move it to the final position at distance r ,

$$V_f = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r}$$

from (1)

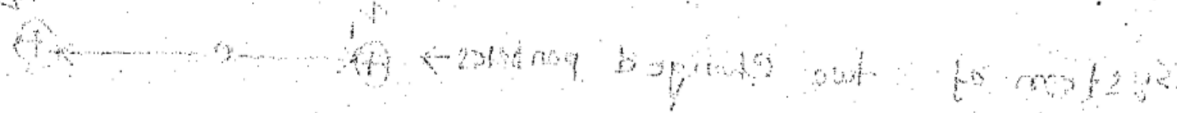
$$U_f = q_1 V_f$$

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

(two-particle system)

Sample problem 24.06: Potential energy of a system

three charged particles.



Particle 1 is fixed in place and particles 2 and 3 move with an initial potential energy U_i for the two-particle system.

As particle 3 moves to its final position and then the system's potential energy is U_f for the three-particle system. The change in potential energy of the system is $\Delta U = U_f - U_i$.

We can relate ΔU to the change in potential energy which we have for particle 3. $U_f = U_i + \Delta U$

$$U_f = U_i + \Delta U \quad \text{where } \Delta U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}}$$

Initially at distance r_{13} , the potential of particle 3 is $V_i = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}}$. When we move it to the final position a distance r_{13}' , $V_f = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}'}$.

$$\Delta U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}'} - \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}}$$

$$U_f = U_i + \Delta U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}'} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

(two-particle system)

$$U_f = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}'} + \frac{q_2 q_3}{r_{23}} \right)$$

Potential of a Charged Isolated Conductor.

① At 23-3 we concluded that $\vec{E} = 0$ for all points inside an isolated conductor. We then used Gauss' law to prove that an excess charge placed on an isolated conductor lies entirely on its surface (also true for empty cavity)

⊗ An excess charge placed on an isolated conductor will distribute itself on the surface of that conductor so that all points of the conductor — whether on surface or inside — come to the same potential. This is true even if the conductor has an internal cavity and even if that cavity contains a net charge.

$$V_j - V_i = - \int_i^j \vec{E} \cdot d\vec{s}$$

Since $\vec{E} = 0$ for all points within a conductor, it follows directly that $V_j = V_i$ for all possible pairs points i and j in the conductor.

